

LLANO, TX, USA

WMO: 720110

Lat: 30.784N

Lon: 98.662W

Elev: 1102

StdP: 14.12

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53983

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.3	28.1	9.8	9.5	38.8	14.1	11.7	41.6	23.3	54.9	20.0	55.1	2.6	0	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.9	102.2	72.2	100.1	72.8	99.1	73.0	79.6	83.1	78.4	85.4	77.2	87.1	8.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	157.7	80.8	76.9	146.0	80.0	74.9	136.5	79.4	44.0	83.2	42.8	88.0	41.4	85.9	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.7	16.5	14.3	DB	17.3	107.0	4.5	1.9	14.0	108.4	11.4	109.5	8.8	110.6	5.6	112.0
			WB	15.5	79.8	4.3	1.3	12.5	80.8	9.9	81.5	7.5	82.3	4.4	83.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.1	49.0	53.6	60.9	67.8	75.0	82.5	84.8	85.7	79.3	69.2	58.2	50.3
	DBStd	14.91	8.87	9.90	8.81	7.68	6.42	3.88	3.55	3.66	5.28	8.27	9.31	8.75
	HDD50	347	125	68	14	2	0	0	0	0	0	2	30	106
	HDD65	1847	501	332	187	59	8	0	0	0	0	56	242	462
	CDD50	6948	93	169	351	535	774	974	1080	1106	878	597	276	115
	CDD65	2974	4	13	60	142	317	524	615	641	429	186	38	5
	CDH74	36314	99	230	668	1545	3271	6468	8086	8715	4781	1941	403	107
	CDH80	19078	17	52	182	573	1448	3499	4717	5231	2483	767	93	16
(14) Wind	WSAvg	5.2	5.0	5.5	6.4	6.5	6.0	5.7	4.9	4.5	4.0	4.5	4.7	4.6
(15) Precipitation	PrecAvg	27.4	1.3	1.9	1.8	2.3	3.5	3.2	1.9	2.2	2.6	2.9	2.1	1.7
	PrecMax	41.6	5.9	6.1	7.0	10.9	7.5	8.9	5.9	11.2	6.8	7.3	9.0	11.4
	PrecMin	16.2	0.0	0.0	0.1	0.2	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.6	1.2	1.5	1.4	1.9	2.1	2.2	1.6	2.2	1.8	1.9	1.8	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.0	86.2	89.9	96.8	99.2	102.4	105.5	106.1	101.9	93.4	88.1	81.8
		MCWB	58.8	64.3	61.4	66.9	72.6	72.8	71.4	72.7	70.4	69.4	69.3	61.4
	2%	DB	76.7	80.8	84.2	90.3	94.6	99.6	101.6	102.3	98.7	90.6	82.1	76.9
		MCWB	58.8	61.1	63.6	67.3	71.9	73.4	71.7	72.6	71.8	70.7	65.2	60.4
	5%	DB	72.0	75.4	81.4	86.3	90.9	97.3	99.6	100.3	95.1	88.1	79.2	72.4
		MCWB	58.1	60.5	63.1	67.3	72.0	73.2	72.9	73.0	72.5	69.7	64.3	60.3
	10%	DB	66.2	72.2	78.5	82.1	88.3	93.5	97.5	99.2	91.4	84.0	74.7	67.9
		MCWB	55.5	59.3	61.9	66.2	71.8	73.2	73.5	73.2	71.9	68.0	63.1	56.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.6	69.0	72.5	75.5	80.7	80.3	80.5	82.1	80.6	77.5	72.4	71.6
		MCDB	71.6	77.0	75.4	82.0	83.6	84.8	82.5	83.4	82.8	85.5	79.6	73.6
	2%	WB	64.5	66.8	68.8	73.2	77.3	78.5	78.7	79.4	78.4	75.3	70.2	66.3
		MCDB	69.7	74.7	76.5	80.6	85.9	87.1	84.5	82.4	85.0	82.3	76.5	72.4
	5%	WB	61.0	64.4	67.1	71.6	75.4	77.1	77.2	77.9	76.5	73.8	68.1	63.0
		MCDB	68.4	70.6	76.0	80.6	83.9	88.1	87.3	86.4	85.9	80.7	75.0	68.0
	10%	WB	56.5	61.8	65.3	70.0	74.2	75.9	76.3	76.6	75.1	72.4	66.1	59.2
		MCDB	66.1	69.1	73.3	78.6	83.4	87.2	87.1	87.4	84.7	79.5	73.1	66.2
(35) Mean Daily Temperature Range	5% DB	MDBR	26.3	26.0	26.2	25.4	22.4	23.0	23.6	24.9	23.6	25.8	26.8	26.3
		MCDBR	35.3	33.7	29.9	30.6	27.2	26.8	27.2	28.5	28.0	27.8	30.2	34.4
		MCWBR	20.1	18.9	13.9	13.5	10.2	6.6	6.1	6.5	8.0	11.6	16.2	20.2
		MCDBR	26.7	25.9	22.5	21.8	21.0	20.4	21.1	22.3	20.5	19.9	22.9	26.0
	5% WB	MCWBR	18.7	17.1	12.7	12.0	9.8	7.0	6.7	7.6	7.9	10.4	14.9	18.9
(40) Clear-Sky Solar Irradiance	taub		0.313	0.319	0.347	0.386	0.421	0.417	0.429	0.419	0.393	0.366	0.336	0.316
	taud		2.518	2.508	2.441	2.354	2.311	2.369	2.377	2.385	2.406	2.468	2.485	2.515
	Ebn at Noon		292	300	297	287	277	276	272	274	279	280	281	285
	Edh at Noon		28	30	35	40	42	39	39	38	36	31	28	26
(44) All-Sky Solar Radiation	RadAvg		995	1189	1503	1801	1884	2158	2149	2014	1662	1354	1041	886
	RadStd		108	177	135	165	196	160	175	118	153	199	133	118

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139

Nomenclature: See separate page